WORKSHEET — UNIT V

IoT Applications

Course: IoT with Applications

Unit: V (3 Hrs / 180 min)

Topics: Smart Agriculture; Healthcare IoT; Industrial IoT; Smart City; Smart Energy and Transportation; Smart Home; The Future of IoT

🎯 Section A — Learning Objectives

1. Explain the common IoT application pipeline — Sensors → Gateway → Cloud → AI → Action — and show how every domain reuses it.

2. Describe IoT in smart agriculture: soil moisture monitoring, automatic irrigation, crop monitoring, weather monitoring and fertilizer management.

3. Describe healthcare IoT: remote patient monitoring (ECG, SpO_2, temperature), smart medical devices and elderly-care alerts.

4. Explain Industrial IoT: machine monitoring, predictive maintenance, asset tracking, industrial automation, digital twins and quality monitoring.

5. Describe IoT in smart cities, smart energy, smart transportation and smart homes.

6. Trace the evolution from IoT 1.0 to 4.0 and plan a mini-project that uses the full pipeline.

🔑 Section B — Key Concepts

No.	Concept	Short Explanation
1	**Common Application Pipeline**	Sensors collect data → a gateway forwards it → the cloud stores and analyses it → AI makes a decision → an action (actuator or alert) follows.
2	**Smart Agriculture**	Soil-moisture, weather and crop sensors drive automatic irrigation, disease detection and fertilizer decisions, saving water and raising yield.
3	**Healthcare IoT**	Wearable and bedside sensors (ECG, SpO_2, temperature) send patient data to clinicians, enabling care at a distance and emergency alerts.
4	**Industrial IoT (IIoT)**	Connected machines and assets in factories support monitoring, automation, predictive maintenance, asset tracking and quality control.
5	**Digital Twin**	A virtual model of a physical machine or system, kept in sync with live sensor data, used to simulate and predict its behaviour.
6	**Smart City**	Sensor networks manage traffic, parking, streetlights, waste, air quality and water so that city services run efficiently.
7	**Smart Energy and Transportation**	Smart meters and the smart grid predict demand and integrate renewables; vehicle tracking and connected vehicles improve fleet efficiency and road safety.
8	**Smart Home**	Connected lighting, security, appliances and voice-controlled devices automate comfort, safety and energy saving in the home.
9	**IoT 1.0 → 4.0 Evolution**	Connected things → cloud-connected → AI-enabled → Edge AI with 5G and digital twins.

📘 Section C — Important Terminology

Keyword	Meaning
Soil Moisture Sensor	A probe that measures the water content of soil and tells the system when to irrigate.

Smart Irrigation	Automatic watering that switches pumps or valves using soil-moisture, weather and crop data.
Remote Patient Monitoring	Tracking a patient's health data outside the hospital with connected devices and alerting clinicians when needed.
ECG	Electrocardiogram: a recording of the heart's electrical activity; wearable ECG sensors help spot irregular heartbeats.
SpO_2	Blood-oxygen saturation, measured by a pulse oximeter; low values can signal breathing problems.
Industrial IoT (IIoT)	IoT applied in factories and plants for machine monitoring, automation and quality control.
Predictive Maintenance	Forecasting equipment failure from sensor data so repairs are planned before a breakdown.
Asset Tracking	Locating and monitoring goods, tools or vehicles in real time using tags, RFID or GPS.
Digital Twin	A virtual replica of a physical machine or system that is updated with live sensor data.
Smart Parking	Sensors detect free and occupied bays and guide drivers to available spaces through an app.
Smart Meter	An electricity, gas or water meter that records usage digitally and reports it automatically.
Smart Grid	An electricity network that uses sensors and two-way communication to balance supply and demand.
Fleet Management	Tracking and managing a group of vehicles — location, routes, fuel use and maintenance — using GPS and IoT.
Intrusion Detection (Home)	Door, window and motion sensors that detect unauthorised entry and trigger alerts.
IoT 4.0	The stage of IoT evolution that combines Edge AI, 5G and digital twins.

✅ Section D — Multiple-Choice Questions [Easy–Medium]

1. Which sequence describes the common IoT application pipeline?

A. Cloud → Sensors → Gateway → Action → AI
B. Sensors → Gateway → Cloud → AI → Action
C. AI → Action → Sensors → Cloud → Gateway
D. Action → Cloud → Gateway → Sensors → AI

2. In smart agriculture, which sensor tells the system when the crops need watering?

A. Smoke sensor
B. Ultrasonic parking sensor
C. Soil moisture sensor
D. Pulse oximeter

3. What does a pulse oximeter measure?

A. Blood-oxygen saturation (SpO_2)
B. The heart's electrical activity
C. Body weight
D. Blood sugar

4. What is the main benefit of emergency alerts in an elderly-care IoT system?

A. Reducing sensor cost
B. Increasing the screen size
C. Storing more videos
D. Notifying caregivers immediately when something goes wrong

5. What is a digital twin?

A. A duplicate sensor installed beside the original
B. A virtual model of a physical asset updated with live sensor data
C. A backup cloud server
D. A second gateway kept for redundancy

6. Which industrial application uses vibration and temperature data to predict machine failure before it happens?

A. Predictive maintenance
B. Smart parking
C. Smart lighting
D. Voice control

7. In a smart city, which service uses sensors to show drivers where free spaces are?

A. Smart waste management
B. Smart streetlights
C. Smart parking
D. Water management

8. What does a smart meter do that a traditional meter does not?

A. Generates electricity
B. Stores energy in a battery
C. Cools the power grid
D. Records usage digitally and reports it automatically

9. Which smart-home function uses door and window sensors and motion detectors to detect unauthorised entry?

A. Smart lighting
B. Intrusion detection
C. Energy management
D. Voice control

10. Which stage of IoT evolution combines Edge AI, 5G and digital twins?

A. IoT 1.0 (connected things)
B. IoT 2.0 (cloud-connected)
C. IoT 3.0 (AI-enabled)
D. IoT 4.0

✏️ Section E — Fill in the Blanks [Easy]

1. In smart agriculture, an automatic _____ system switches the pump on when soil moisture falls below a set level.

2. An _____ sensor records the electrical activity of the heart.

3. Locating and monitoring goods, tools or vehicles in real time is called _____ tracking.

4. A _____ twin is a virtual replica of a physical machine that is updated with live sensor data.

5. _____ management uses GPS and sensors to monitor the location, routes and fuel use of a group of vehicles.

☑️ Section F — True or False [Easy]

1. The pipeline Sensors → Gateway → Cloud → AI → Action can be reused across many different IoT application domains. (True / False)

2. Smart irrigation always waters crops on a fixed timetable, whatever the soil moisture. (True / False)

3. A pulse oximeter measures blood-oxygen saturation (SpO_2). (True / False)

4. Smart meters can only be read manually by a technician. (True / False)

5. Intelligent traffic systems can adjust signal timings using live traffic data. (True / False)

🔗 Section G — Match the Following [Medium]

Column A	Column B (shuffled)
1. Soil moisture sensor	a. Virtual replica of a machine updated with live sensor data
2. SpO_2	b. Blood-oxygen level measured by a pulse oximeter
3. Digital twin	c. Reports electricity usage automatically in near real time
4. Smart parking	d. Detects free bays and guides drivers to them through an app
5. Smart meter	e. Tells the system when crops need watering
6. Fleet management	f. Door, window and motion sensors that detect unauthorised entry
7. Intrusion detection	g. Tracking the location, routes and fuel use of vehicles

📝 Section H — Short-Answer Questions [Medium]

1. Write the common IoT application pipeline and explain what happens at each stage.

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2. How does smart irrigation save water?

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3. How does remote patient monitoring benefit patients and doctors?

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4. What is a digital twin, and how is it used in industry?

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5. Describe the four stages of IoT evolution from IoT 1.0 to IoT 4.0.

📄 Section I — Descriptive Questions [Medium–Advanced]

1. Explain smart agriculture using the pipeline Sensors → IoT Gateway → Cloud → AI → Irrigation, covering soil moisture monitoring, automatic irrigation, crop monitoring and disease detection, weather monitoring and fertilizer management.

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2. Describe healthcare IoT: remote patient monitoring, ECG, SpO_2 and temperature monitoring, smart medical devices, and elderly-care systems with emergency alerts.

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3. Explain Industrial IoT — machine monitoring, predictive maintenance, asset tracking, industrial automation, digital twins and quality monitoring — with an example of each.

4. Describe IoT applications in smart cities, smart energy and transportation, and smart homes, and show how each reuses the common pipeline.

💻 Section J — Practical / Application-Based Questions [Advanced]

Practical 1

Problem: Write a Python function that implements a smart-irrigation rule.

Task:

1. Define irrigation_decision(moisture_pct, rain_expected).

2. Return “PUMP ON” when soil moisture is below 30 and no rain is expected; otherwise return “PUMP OFF”.

3. Test it with the inputs (22, False), (22, True) and (45, False) and print each result.

Expected Output: *Three lines: PUMP ON, PUMP OFF, PUMP OFF.*

Practical 2

Problem: Design the sensing and action stages for three smart-city services.

Task:

1. Smart parking: choose a sensor and describe what the driver’s app shows.

2. Smart streetlights: choose the sensors and state when the light should dim or brighten.

3. Smart waste management: choose a sensor and state what alert is sent and to whom.

Expected Output: *A table giving the sensor, the data sent to the cloud and the resulting action for each service.*

Practical 3

Problem: Process hourly smart-meter readings to find total energy use, its cost and the peak hour.

Task:

1. Store the readings [1.2, 0.8, 0.6, 2.5, 3.1, 1.9] (in kWh) in a list.

2. Compute the total consumption and its cost at 7 rupees per kWh.

3. Find the highest reading and its position in the list (counting from 0).

Expected Output: *Total 10.1 kWh, cost 70.7 rupees, and a peak of 3.1 kWh at index 4.*

🗺️ Section K — Concept-Mapping Activity

IOT APPLICATIONS

Complete the map below: fill in the missing sub-topic name or example for each branch.

Branch 1	Branch 2	Branch 3	Branch 4
Smart Agriculture	**Healthcare IoT**	**Industry and City**	**Energy, Transport and Home**
Example: _____	Example: _____	Example: _____	Example: _____

(Answer key — example values: soil moisture, automatic irrigation, crop disease detection; ECG, SpO_2, emergency alerts; predictive maintenance, digital twins, smart parking; smart meters, fleet management, smart lighting)

🏢 Section L — Case Study / Real-World Activity

Scenario

A university wants to become a "smart campus". It plans to (a) switch streetlights on only when people or vehicles are nearby, (b) show free parking spaces on a mobile app, (c) alert staff when waste bins are nearly full, and (d) monitor the energy use of each building. Sensors across the campus send data through gateways to a cloud platform, where analytics produce dashboards and alerts.

Questions

1. Name a suitable sensor for each of the four services.

2. Map the smart-parking service onto the pipeline Sensors → Gateway → Cloud → AI → Action.

3. Which long-range technology from Unit II would suit the bins and streetlights spread across the campus, and which task would you process at the edge?

4. Suggest one security measure from Unit IV for this system and one expected benefit of the smart campus.

Expected Learning Outcome

Students apply the common IoT pipeline, and ideas from earlier units, to plan a multi-service smart-campus project.

📊 Section M — Self-Assessment

☐ **Yes** ☐ **Partly** ☐ **No** I can explain the major concepts of this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can define the important terminology from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can solve problems/exercises from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can apply the concepts of this unit to a new situation.

💭 Section N — Student Reflection

What did I learn in this unit?

What was difficult for me?

What do I need to practice more?

One important concept I learned:

🗝 Section O — Instructor Answer Key

(For instructor use — covers all objective and subjective items in this worksheet.)

MCQ Answers

- 1. B — Data flows from sensors through a gateway to the cloud, where AI decides and an action follows.
- 2. C — A soil moisture sensor measures soil water content, which triggers irrigation.
- 3. A — A pulse oximeter reports blood-oxygen saturation (SpO_2); ECG measures the heart's electrical activity.
- 4. D — Emergency alerts notify caregivers at once, so help can be given quickly.
- 5. B — A digital twin is a virtual model of a physical asset kept in sync with live sensor data.
- 6. A — Predictive maintenance uses vibration, temperature and similar data to forecast failure.
- 7. C — Smart parking sensors detect free bays and show them to drivers in an app.
- 8. D — A smart meter records usage digitally and reports it automatically, enabling near real-time data.
- 9. B — Door and window sensors plus motion detectors form the intrusion-detection function.
- 10. D — IoT 4.0 combines Edge AI, 5G and digital twins.

Fill in the Blanks — Answers

- 1. irrigation
- 2. ECG (electrocardiogram)
- 3. asset
- 4. digital
- 5. Fleet

True/False — Answers

- 1. True
- 2. False
- 3. True
- 4. False
- 5. True

Match the Following — Answer Key

1 → e 2 → b 3 → a 4 → d 5 → c 6 → g 7 → f

Short Answer — Key Points

- 1. Sensors collect data from the physical world; the gateway gathers and forwards it; the cloud stores and analyses it; AI turns the analysis into a decision or prediction; the action stage switches an actuator or sends an alert (e.g. a pump, valve or notification).
- 2. Sensors measure the actual soil moisture and weather data, so water is applied only when and where it is needed, instead of on a fixed schedule, reducing waste and over-watering.
- 3. Patients can be monitored at home through ECG, SpO_2 and temperature sensors, saving hospital visits; doctors receive continuous data and alerts, so they can spot problems early and respond faster.
- 4. A digital twin is a virtual model of a machine or plant fed with live sensor data. Engineers use it to monitor condition, test "what-if" changes safely and predict failures before they happen.
- 5. IoT 1.0: connected things; IoT 2.0: cloud-connected devices; IoT 3.0: AI-enabled analytics and decisions; IoT 4.0: Edge AI combined with 5G and digital twins.

Descriptive Questions — Key Points

- 1. Pipeline: soil, temperature, humidity and rain sensors → IoT gateway → cloud → AI → irrigation. Soil moisture monitoring shows when to water; automatic irrigation switches the pump; crop monitoring uses cameras or drones with computer vision for disease detection; weather monitoring avoids watering before rain; smart fertilizer management uses soil nutrient data to apply only what is needed.
- 2. Remote patient monitoring sends vital signs to clinicians. ECG sensors track heart rhythm, SpO_2 sensors track blood-oxygen level and temperature sensors detect fever. Smart medical devices include connected glucose monitors and insulin pumps. Elderly-care systems use wearables or room sensors to detect falls or inactivity and send emergency alerts to caregivers.
- 3. Machine monitoring (vibration, temperature, current); predictive maintenance (forecast failure); asset tracking (RFID/GPS for tools and goods); industrial automation (sensors and controllers running processes automatically); digital twins (virtual model for simulation); quality monitoring (cameras or sensors detect defects). Give one example for each.
- 4. Smart city: traffic, parking, streetlights, waste, air quality and water management. Smart energy: smart meters, smart grid, demand prediction, renewable-energy and building energy monitoring. Smart transportation: vehicle tracking, fleet management, intelligent traffic, connected vehicles, accident detection. Smart home: lighting, security, appliances, energy management, voice control, intrusion detection. Each follows sensors → gateway → cloud → AI → action.

Practical Activities — Expected Outputs

- 1. Sample logic: def irrigation_decision(moisture_pct, rain_expected): return “PUMP ON” if moisture_pct < 30 and not rain_expected else “PUMP OFF”. The tests print PUMP ON (dry, no rain), PUMP OFF (dry but rain expected) and PUMP OFF (soil already moist).
- 2. (1) Smart parking: ultrasonic, magnetic or infrared bay sensor (or a camera) → occupancy sent to the cloud → the app shows the number and location of free bays. (2) Smart streetlights: a light (LDR) sensor plus a motion/PIR sensor → dim the light when it is bright or nobody is near, brighten when someone approaches at night. (3) Smart waste: an ultrasonic fill-level sensor → alert sent to the collection staff when the bin is nearly full, so routes are planned efficiently.
- 3. Sample logic: kwh = [1.2, 0.8, 0.6, 2.5, 3.1, 1.9]; total = round(sum(kwh), 2) gives 10.1; cost = round(total * 7, 2) gives 70.7; peak = max(kwh) gives 3.1 and kwh.index(peak) gives 4.

Concept Map — Sample Answers

- Smart Agriculture: soil moisture monitoring, automatic irrigation, crop disease detection, weather monitoring, fertilizer management. Healthcare IoT: ECG, SpO_2, temperature, smart medical devices, emergency alerts. Industry and City: predictive maintenance, asset tracking, digital twins, smart parking, smart waste. Energy, Transport and Home: smart meters, fleet management, connected vehicles, smart lighting, intrusion detection.

Case Study — Key Points

- 1. Streetlights: light (LDR) and motion/PIR sensors. Parking: ultrasonic, magnetic or infrared bay sensors or cameras. Waste bins: ultrasonic fill-level sensors. Building energy: smart meters or current sensors.
- 2. Bay sensors detect occupancy → gateway collects the readings → cloud stores them and builds a live map → AI predicts peak parking demand → action: the app shows free bays and guides drivers.
- 3. LoRaWAN (long range, low power, battery-friendly, tiny payloads) suits bins and streetlight sensors. Edge processing: streetlight dimming based on local motion, which needs an instant response and no cloud round trip.
- 4. Security: encrypted communication (TLS/DTLS), unique device credentials and signed firmware updates. Benefits: lower energy use, easier parking, efficient waste collection and better campus-wide visibility.